Team Number: 44

Project Title: Breast Cancer Stage Classification with miRNA

Report Date: 8/25/25

Part one: Progress status as a team

All team members have detailed tasks listed on Teams Planner: YES or NO; if NO, explain.

Yes

Please answer the following questions:

1. Have you met as a team this past week? If yes, give date/time, and the members attended the meeting. If no, explain.

   We did not hold a formal team meeting; however, we coordinated daily through our team's Discord server to discuss progress, assign tasks, and resolve questions/issues.

2. Have you met with the sponsor as a team? If yes, give date/time, and the member attended the meeting. If no, explain.

   Yes. The entire team (Joshua Adegboyo, Anika Baro Villa, Nicholas Campos, Brehon Parker) met with Dr. Ali Ibrahim on Monday, August 25, 2025, at 9 p.m.

3. Describe verbally the tasks completed the past week and the challenges faced.

   Last week (Week 2), we had our first meeting of this phase with Dr. Ali Ibrahim. During the meeting, we were assigned some initial tasks to move forward with the project. These included reviewing a research article closely related to our own project, downloading breast cancer miRNA expression and metadata data from the Genomic Data Commons (GDC) Portal, and setting up our working environment using Jupyter notebooks. We also discussed and defined the tasks we will be working on in the upcoming weeks.

   In addition to working on the tasks assigned by Dr. Ali, we revised the project proposal by drafting new sections, which included a 'Summary of Completed Tasks and Sub-Tasks', a 'Detailed Task Breakdown with Member Responsibilities', a set of 'Use-Case Scenarios and Test Cases', and a revised version of our Work Breakdown Structure (WBS). Also, we created shared folders for ED2 deliverables within Microsoft Teams and updated our project planner. For each team member, we mapped out approximately 14 weeks' worth of tasks leading to the project's completion.

   The most challenging aspect of the week was creating our individual lists of detailed tasks and subtasks, as we anticipate that several of these will evolve as the project progresses.

4. Describe the tasks to be completed the coming week.

   This coming week (Week 3), we will focus on cleaning and labeling the miRNA expression dataset. First, we will clean the raw expression files by checking their structure and integrity, applying a $\log_2(x+1)$ transformation for normalization, handling missing values, and removing any obvious outliers. Once the data has been cleaned, we will then link it with the clinical metadata by matching case identifiers, allowing us to assign each sample the appropriate breast cancer stage label (I–IV). Dr. Ali Ibrahim will guide us through each of these steps to ensure we are following the correct procedures and maintaining consistency with project goals.

Current Report Summary. Date: 8/25/25

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
Joshua Adegboyo	2	0	2
Anika Baro Villa	3	0	2
Nicholas Campos	2	0	1
Brehon Parker	3	0	2

Name of the Member: Joshua Adegboyo

Report Date: 8/25/25

Task 2.1: Led team meeting with Project Sponsor (8/25/25)

The goal of this meeting was to go over our project and upcoming tasks with our project sponsor. He walked us through downloading the miRNA and clinical metadata from the GDC Portal so we can begin on the data cleaning upon our next meeting (September 1st, 2025) and showed us various feature selection methods we would be using for the upcoming tasks. The total file count we would be working with is 1,209 files across 1,081 cases.

Task 2.2: Reviewed and modified revised project proposal (8/30/25)

For this task, I revised the project proposal by drafting new sections which included a "Summary of Completed Tasks and Sub-Tasks," which outlines the progress we have made so far; a "Detailed Task Breakdown with Member Responsibilities," aligning specific tasks and subtasks with individual team members based on the WBS; a set of "Use-Case Scenarios and Test Cases" that covers key system interactions, such as uploading miRNA files, initiating predictions, and viewing the resulting classifications; and a revised WBS that reflects our current timelines and responsibilities, which continue to evolve as the project progresses.

Task 3.1: Clean miRNA expression files (9/7/25)

With Dr.Ali's guidance, we would begin the data cleaning procedures to ensure the raw miRNA expression files are consistent, complete, and properly normalized. This would include checking file integrity, handling missing or invalid values, applying a $\log_2(x+1)$ transformation for normalization, and removing outliers. The expected outcome would be a clean, well-structured dataset prepared for integration with clinical metadata in the next task.

Task 3.2: Link cleaned miRNA expression data with clinical metadata by case IDs (9/7/25)

In collaboration with Dr. Ali, the team would link the cleaned miRNA expression dataset with the corresponding clinical metadata using unique case identifiers. This process would involve aligning sample IDs across both datasets, verifying the presence and accuracy of cancer stage labels (I–IV), and addressing any inconsistencies or missing entries. The team would also assess the distribution of samples across stages to understand class balance. The final output would be a fully labeled dataset, combining miRNA features with clinical information, prepared for downstream tasks such as feature selection and model training.

Name of the Member: Anika Baro Villa

Report Date: 8/25/25

Task 2.1: Download data (miRNA + clinical) from GDC Portal (8/26/25)

The goal of this task was to download the miRNA and clinical metadata from the GDC Portal so we can proceed with data cleaning and labeling. Following Dr. Ali Ibrahim's instruction, I accessed the GDC Portal and applied various filters to isolate the relevant data to our project: primary site as "breast," experimental strategy as "microRNA sequencing," data type as "miRNA expression quantification," and file format as .txt. I also ensured the access level was set to open and included both normal and tumor tissue types. In addition to the expression data, I selected clinical metadata in JSON format for the matched cases. Additionally, I confirmed that the total file count (1,209 files across 1,081 cases) matched the figures previously provided by Dr. Ali, added the selected items to the cart and downloaded the manifest, cart, and clinical data files.

Task 2.2: Write new sections for the revised project proposal (8/30/25)

For this task, I helped revise the project proposal by drafting several new sections. These include a "Summary of Completed Tasks and Sub-Tasks," which outlines the progress we have made so far; a "Detailed Task Breakdown with Member Responsibilities," aligning specific tasks and subtasks with individual team members based on the WBS; a set of "Use-Case Scenarios and Test Cases" that covers key system interactions, such as uploading miRNA files, initiating predictions, and viewing the resulting classifications; and a revised WBS that reflects our current timelines and responsibilities, which continue to evolve as the project progresses.

Task 2.3: Create shared folders for ED2 deliverables and update project planner (8/30/25)

I collaborated with the team to create shared folders within Teams dedicated to ED2 deliverables and progress reports. Additionally, I added tasks to my assigned bucket in the project planner, outlining my responsibilities for the remaining 14 weeks of the project. These tasks are aligned with the updated WBS to support weekly tracking of task completion and overall project progress.

Task 3.1: Clean miRNA expression files (9/7/25)

Following Dr. Ali's guidance, the team would carry out data cleaning procedures to ensure the raw miRNA expression files are consistent, complete, and properly normalized. This would include checking file integrity, handling missing or invalid values, applying a $\log_2(x+1)$ transformation for normalization, and removing outliers. The expected outcome would be a clean, well-structured dataset prepared for integration with clinical metadata in the next task.

Task 3.2: Link cleaned miRNA expression data with clinical metadata by case IDs (9/7/25)

In collaboration with Dr. Ali, the team would link the cleaned miRNA expression dataset with the corresponding clinical metadata using unique case identifiers. This process would involve aligning sample IDs across both datasets, verifying the presence and accuracy of cancer stage labels (I–IV), and addressing any inconsistencies or missing entries. The team would also assess the distribution of samples across stages to understand class balance. The final output would be a fully labeled dataset, combining miRNA features with clinical information, prepared for downstream tasks such as feature selection and model training.

Name of the Member: Nicholas Campos

Report Date: 8/25/25

Task 2.1: Download data miRNA and clinical from GDC Portal (8/26/25)

My objective for this task was to acquire the necessary miRNA expression and clinical data from the GDC Portal to enable subsequent data processing phases. Under Dr. Ali Ibrahim's guidance, I navigated the GDC Portal and configured specific filtering parameters for our breast cancer research: setting the primary site to "breast," selecting "microRNA sequencing" as the experimental strategy, choosing "miRNA expression quantification" as the data type, and specifying .txt file format. For clinical information, I obtained the corresponding metadata in JSON format. I verified that our data acquisition matched the expected counts of 1,209 files spanning 1,081 cases as previously outlined by Dr. Ali, then proceeded to add all selected files to the cart and successfully downloaded the manifest, cart contents, and clinical data files.

Task 2.2: Revised project proposal (8/31/25)

My contribution to the project proposal revision involved me creating my deliverable and milestones section. This required developing a comprehensive breakdown of specific project outcomes, timeline expectations, and key checkpoint markers that align with our overall system development goals. I established clear deliverable specifications for each major project phase, defined measurable milestone criteria to track our progress effectively, and coordinated these elements with our team's individual responsibilities to ensure proper accountability and timeline adherence throughout the remaining project duration.

Task 3.1: Begin cleaning data (9/7/25)

Under Dr. Ali's direction, our team will implement comprehensive data preprocessing protocols to ensure miRNA expression files meet quality standards for analysis. This process will involve validating file formats and completeness, addressing missing or corrupted data points, implementing $\log_2(x+1)$ normalization transformations, and identifying and removing statistical outliers. The deliverable will be a standardized, analysis-ready dataset suitable for clinical metadata integration.

Name of the Member: **Brehon Parker**

Report Date: **8/25/25**

Task 2.1: Download data (miRNA + clinical) from GDC Portal (8/26/25)

As part of this task, I coordinated our meeting with Dr. Ali Ibrahim and also performed the dataset download from the GDC Portal. Following his instructions, I applied the filters for breast cancer (primary site = breast, experimental strategy = microRNA sequencing, data type = miRNA expression quantification, .txt format, open access) and included the corresponding clinical metadata in JSON format. After downloading the manifest and clinical files, I verified that the counts matched the expected totals (1,209 files across 1,081 cases).

Task 2.2: Write new sections for the revised project proposal (8/30/25)

For this task, I updated my portion of the project proposal. I listed my weekly responsibilities in detail, broke them into subtasks, and aligned them with the milestones in the Work Breakdown Structure (WBS). I also checked that the timeline of my contributions matched the project plan the team created. While the scope of my work was focused on my own role, documenting it in the proposal helped the team ensure that all members' responsibilities were clearly represented and tied back to deliverables.

Task 2.3: Create shared folders for ED2 deliverables and update project planner (8/30/25)

For this task, I worked with the team to coordinate who would create the shared folders in Microsoft Teams for our deliverables. The goal was to have a central space where we could store reports, figures, and progress documents. After sthe folders were set up, I updated my tasks in the Teams Planner by adding items to my assigned bucket. These tasks reflect my responsibilities over the remaining weeks of the project and align with the updated WBS. This setup makes it easier for the team to track progress and ensures each member knows where to place their work.

Task 3.1: Clean miRNA expression files (9/7/25)

This week we will begin cleaning the raw miRNA expression files, working closely with Dr. Ali. The goal is to make sure the data is consistent and ready for analysis. With Dr. Ali's suggestion, we will check that the files load properly, apply the $\log_2(x+1)$ transformation to normalize the values, and look for issues like missing entries or outliers. We'll also help keep notes on what steps we took so we can repeat the same process across all files. This task will prepare us to connect the cleaned data to the clinical information in the next step.

Task 3.2: Link cleaned miRNA expression data with clinical metadata by case IDs (9/7/25)

Once the cleaning step is finished, we'll work with Dr. Ali to link the expression dataset to the clinical metadata. I will assist double-checking that the sample IDs align correctly across the two files and help identify any missing or mismatched entries. I'll also review the stage labels (I–IV) to make sure they look accurate and balanced. Dr. Ali will be guiding us through this process so that we follow the proper approach and keep the dataset reliable for later tasks like feature selection and model training.